THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapter 11: **Keeping Time with the Skies** | Class 8 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** rotation of the Earth on its axis	Earth spins from west to east.
2	**(a)** 27 days	Orbital period of the Moon.
3	**(b)** solar calendar	Gregorian.
4	**(b)** 24 hours	One day.
5	**(d)** 29.5 days	Synodic period.
6	**(b)** lunar calendar	Months follow the Moon.
7	**(d)** 365 and a quarter days	One year.
8	**(b)** full moon	Purnima.
9	**(c)** 365 days (366 in a leap year)	Solar.
10	**(d)** 366 days	An extra day is added to February.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Earth rotates from west to east (1); So the Sun appears to move east to west (1).	1+1
12	Sun is at its highest point in the sky (1); Its rays fall almost vertically (1).	1+1
13	Solar calendar (1); Lunar calendar (1).	1+1
14	Rotation of the Earth on its axis (1); Half the Earth faces the Sun at a time (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Fix a vertical stick on level ground in the Sun (1); Mark the end of its shadow every hour (1); Shadow positions show the time of the day (1).	1+1+1
16	Earth's axis is tilted by 23.5 degrees (1); The tilt keeps the same direction as Earth revolves (1); Sun's rays fall differently, changing day length (1).	1+1+1
17	Day: one rotation of Earth (1); Month: one phase cycle of the Moon (1); Year: one revolution around the Sun (1).	1+1+1
	OR: Morning: long, towards west (1); Noon: shortest (1); Evening: long, towards east (1).	1+1+1
18	Day and night (1); Apparent motion of the Sun and stars (1); Time zones (1).	1+1+1
	OR: Observatory built by Sawai Jai Singh II (1); Large stone instruments such as Samrat Yantra (1); Used to measure time and track the Sun and stars (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Rotation: day and night (1); Revolution: year (1); Axis tilt 23.5 degrees (1); Seasons (1); Day length changes (1).	1+1+1+1+1
	OR: Sunlit half always faces the Sun (1); Moon orbits the Earth (1); Angle between Sun and Moon changes (1); We see varying fraction of the lit half (1); Full moon when Earth is between (1).	1+1+1+1+1
20	New moon (1); Waxing crescent, first quarter, gibbous (1); Full moon (1); Waning phases (1); Reflected sunlight and relative positions (1).	1+1+1+1+1
	OR: Sundial uses a gnomon (1); Shadow shows time (1); Samrat Yantra is a giant sundial (1); Gives accurate time to seconds (1); Built in Jaipur in 1734 (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 11: **Keeping Time with the Skies** | Class 8 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** new moon	Amavasya.
2	**(b)** 354 days	12 x 29.5.
3	**(b)** February	29 February.
4	**(c)** the changing positions of the Moon, Earth and Sun	We see different parts of the lit half.
5	**(a)** lunisolar	Indian calendars.
6	**(c)** sundial	Gnomon casts the shadow.
7	**(b)** reflects sunlight	Not luminous.
8	**(b)** tilt of the Earth's axis and its revolution	Tilt of 23.5 degrees.
9	**(a)** noon	Sun is highest.
10	**(d)** its rotation period equals its orbital period	Tidal locking.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Rotation: spinning on its axis, 24 hours (1); Revolution: orbit around the Sun, one year (1).	1+1
12	A star very near the north celestial pole (1); It shows the north direction (1).	1+1
13	A year with 366 days (1); Earth takes about 365.25 days to revolve (1).	1+1
14	Different shapes of the bright part of the Moon seen from Earth (1); From new moon to full moon and back (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Earth takes 365.25 days to go round the Sun (1); Four times 0.25 = 1 day (1); The extra day is added to February (1).	1+1+1
16	Rise in the east and set in the west (1); Star trails in a long exposure (1); Pole Star appears fixed (1).	1+1+1
17	They follow the lunar or lunisolar calendar (1); Lunar year is shorter than the solar year (1); An extra month is added from time to time (1).	1+1+1
	OR: New moon: not visible (1); First quarter and full moon: waxing (1); Last quarter: waning (1).	1+1+1
18	Find the two pointer stars at the end of the cup (1); Extend the line about five times their gap (1); It points to the Pole Star (1).	1+1+1
	OR: Moon takes about half the lunar cycle to go from new to full (1); Total cycle about 29.5 days (1); Half is about 15 days (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Locate the Big Dipper (1); Use pointer stars (1); Pole Star lies above Earth's axis (1); Other stars appear to circle around it (1); North direction found (1).	1+1+1+1+1
	OR: Sun rises in the east (1); Star trails (1); Time zones differ (1); Foucault pendulum rotates (1); Day and night alternate (1).	1+1+1+1+1
20	Solar: Gregorian, 365/366 days (1); Lunar: Islamic, about 354 days (1); Lunisolar: Indian calendars (1); Leap year adjustments (1); Extra month in lunisolar (1).	1+1+1+1+1
	OR: Months start from new or full moon (1); Festivals fixed by lunar days (tithi) (1); Seasons need solar corrections (1); Adhik maas (1); Examples (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

ANSWER KEY & MARKING SCHEME SET 3

Chapter 11: **Keeping Time with the Skies** | Class 8 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** 23.5 degrees	From the vertical.
2	**(d)** west	Sun is in the east.
3	**(a)** two weeks	Waxing phase.
4	**(c)** 21 June	Summer solstice.
5	**(b)** stay almost fixed in the northern sky	It lies near the north celestial pole.
6	**(b)** Sun	Sunlit half.
7	**(b)** 21 March and 23 September	Equinoxes.
8	**(c)** rotation of the Earth	Apparent motion.
9	**(c)** increasing	Growing.
10	**(d)** 22 December	Winter solstice.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	About 29.5 days (2).	2
12	A vertical stick whose shadow shows time (1); Used in sundials (1).	1+1
13	The Moon rotates once in the time it takes to go round the Earth (1); Same face points to Earth (1).	1+1
14	Solstices: longest and shortest days (1); Equinoxes: day and night equal (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Moon reflects sunlight (1); Only half is lit (1); We see different portions of the lit half as it moves (1).	1+1+1
16	Earth's axis is tilted (1); Sun's rays fall differently in different seasons (1); Day is longest at summer solstice (1).	1+1+1
17	Equator: about 12 hours all year (1); Higher latitudes: longer days in summer (1); Higher latitudes: shorter days in winter (1).	1+1+1
	OR: Solar based on the Sun, lunar on the Moon (1); Solar year about 365 days; lunar year about 354 days (1); Lunar needs adjustment (1).	1+1+1
18	Plan festivals and holidays (1); Agriculture and seasons (1); Records of events (1).	1+1+1
	OR: Shadow of a stick (1); Position of the Sun at sunrise (1); Rising of particular stars (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Morning shadow long to the west (1); Noon shortest (1); Evening long to the east (1); Shadow shorter in summer at noon (1); Longer in winter (1).	1+1+1+1+1
	OR: Earth's revolution changes night sky (1); Orion in winter (1); Ursa Major (Saptarishi) (1); Cassiopeia (1); Pole Star remains near north (1).	1+1+1+1+1
20	21 June: longest day (north) (1); 22 December: shortest day (1); 21 March: equinox (1); 23 September: equinox (1); Day equals night at equinox (1).	1+1+1+1+1
	OR: Moon between Sun and Earth: solar eclipse (1); Earth between Sun and Moon: lunar eclipse (1); Moon's orbit is tilted by about 5 degrees (1); They line up only occasionally (1); Never look at the Sun directly (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.